Multi-Knowledge Augmented Preference-Optimized GAN for Scalable and High-Fidelity Human-Driven Text-to-Image Synthesis

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ABSTRACT

Text-to-image (T2I) synthesis can be controlled by humans and produce images that accurately reflect the visual identity of a reference human and follows textual cues. Although the Knowledge and Multi-Detail Enhanced GAN (Xu et al., 2025) was the first to propose an efficient GAN-based solution that relies on external knowledge and CLIP multi-layer properties, several gaps exist: (1) fixed, single-image knowledge retrieval using a fixed training set impedes open-domain generalization and diversity; (2) complex multi-human interactions, poses, and sem We present MK-POGAN, a Multi-Knowledge Augmented Preference-Optimized GAN, a Multi-Knowledge Dynamic Retrieval, a Cross-Modal Multi-Knowledge Fusion Transformer, an Adaptive Pose-Aware Detail Enhancer and the incorporation of rich human feedback through Reward Preference Optimization (RPO). The significant experiments on extended FFHQ, COCO-Human, and a novel Open-Human dataset achieve better identity conservation (↑12-18% over baselines), prompt consistency (↑28%), and FID scores, and require only slightly less than 300x inference compared to diffusion models and less than 6% parameters. Our technique can render 1024x1024 natively and in open-vocabulary, multi-subject situations.

Index Terms- Human-controlled text-to-image synthesis, Generative Adversarial Networks, Knowledge augmentation, Preference optimization, Multi-detail preservation.

I. INTRODUCTION

Traditional text-to-image synthesis produces semantically consistent images given only descriptions of the text. The human-driven variants introduce a reference human image to impose identity preservation, which facilitates applications in personalized advertising, virtual try-ons, creative content, and online avatars [1], [2], [3].

Diffusion-based models (e.g., DreamBooth, IP-Adapter) have high quality, but are expensive to compute, slow to infer, and produce unnatural collage-like results when using humans as references and backgrounds. These were tackled with the Knowledge and Multi-Detail Enhanced GAN (Xu et al., 2025) through external knowledge retrieval and CLIP multi-layer capabilities, with efficiency improvements (300× faster, 5% parameters) on FFHQ faces [4].

A. Gaps and Motivations Identified.

Although it has strengths, previous research indicates some shortcomings. To begin with, top-1 similarity based on a training subset (20% of FFHQ) is involved in both static and single knowledge retrieval, which limits diversity and cannot work in open-domain conditions with novel scenes or multi-subject interactions. Second, sparse detail and pose control multi-layer CLIP benefits, but do not explicitly represent pose, clothing, or multi-human spatial relationships, resulting in identity blending with complex prompts. Third, fixed 224×224 generation with external super-resolution limits high-fidelity outputs with no native support for 1024+ resolution. Fourth, without human preference alignment, semantic harmony is suboptimal in ambiguous cases [5]. Fifth, this dataset is mostly face-centric (FFHQ), and thus it cannot be generalized to full-body or real-life open-vocabulary.

MK-POGAN fills these gaps inspired by recent progress in StyleGAN-T [6], GALIP [7], and preference-based RLHF to T2I [5], [8]. We are major contributors to:

1. Dynamic Multi-Knowledge Retrieval Module: Retrieves and fuses k semantically-relevant images of a large external database with hierarchical CLIP embeddings.

2. Cross-Modal Multi-Knowledge Fusion Transformer (CMK-FT): New attention mechanism, which adapts to weight knowledge regions and also solves text-knowledge conflicts.

3. Adaptive Pose-Aware Multi-Detail Enhancer (AP-MDE): CLIP multi-layer features are extended with pose estimation (OpenPose) and transformer-based fusion to achieve strong identity and structural preservation.

4. Preference-Optimized Training with RP: Adds rich human feedback (RichHF-18K style) with a λ-Harmonic reward to achieve faster convergence and improved semantic alignment.

5. Global Evaluation: New state-of-the-art on FFHQ+, COCO-Human and Open-Human datasets, including native 10241024 generation and ablation experiments.

Our approach is entirely GAN-based and efficient, yet attains diffusion-level quality, enabling practicable, open-domain creative tools.

II. RELATED WORK

A. Text-to-Image GANs and Diffusion Models.

Early GAN-based T2I systems [9], [10] relied on attentional mechanisms, but had poor resolution and mode collapse. New developments such as GALIP [7] and StyleGAN-T [6] use CLIP to achieve more alignment and scalability. Diffusion models [11], [12] are the best in quality but computationally expensive, with billions of parameters and minutes per sample, which cannot be used in practice in real-time or on edges.

B. Human-Driven (Subject-Driven) T2I.

Approaches can be divided into fine-tuning (Textual Inversion [13], DreamBooth [1], Custom Diffusion [2]) or encoder-based (FastComposer [3], IP-Adapter [14]). These are very effective in identity preservation and need per-subject optimization or are not efficient and natural. More recent preference optimization [8] and rich human feedback methods [5] are better at achieving this alignment, but are not combined into an efficient GAN-based framework.

C. Gap Addressed

The previous GAN models do not have a dynamically operating external multi-knowledge retrieval and explicit preference guidance towards open-domain human customization. There is no known approach to combine efficiency of GAN with large-scale knowledge augmentation, explicit pose modeling and human preference learning at the same time - the key contribution of the work.

III. PROPOSED METHOD

A. Overall Pipeline

MK-POGAN takes place in 5 steps given a text prompt t and a reference human image h: (1) dynamically retrieve k=3 knowledge images {k_i} in a large-scale external database; (2) generate multi-knowledge augmented input by combining retrieved images with the reference; (3) initial feature extraction using CLIP-ViT encoder with noise injection to achieve stochasticity; (4)

B. Dynamic Multi-Knowledge Retrieval

The CLIP pre-embeddings are taken out of a huge external database based on a LAION-5B subset. The similarity score of each candidate knowledge image k i to a fixed reference h and text prompt t is calculated as:

$$\alpha_i = \mathrm{softmax}(F_k_i \cdot t + \lambda \cdot \mathrm{pose_sim}(h, k_i))$$

Weighted averaging over the CLIP feature maps of the top-k candidates is used to select the top-k candidates and fuse them. This is a dynamic solution as opposed to the previous methods of static retrieval which fix a subset of training allowing the generalization of the content to new scenes and multi-subject setups.

C. Cross-Modal Multi-Knowledge Fusion Transformer (CMK-FT)

The CMK-FT uses multi-head cross-attention to combine knowledge features Fk, original image features Fin and text embeddings

$$:\mathrm{Attn}(Q, K, V) = \mathrm{softmax}(QK^T / \sqrt{d_k})\ V$$

$$Q = W_q\ [F_{in};\ t],\quad K,V = W_{\{k/v\}}\ F_k$$

Semantically inconsistent knowledge areas are filtered by a conflict mask M = 0(cos(F k, t) > 0) to avoid noisy semantic data that is caused by retrieved images that are incompatible with the text prompt. This process is essential in the open-vocabulary environment in which knowledge images can be partly coincident with the scene of interest.

D. Adaptive Pose-Aware Multi-Detail Enhancer (AP-MDE)

AP-MDE is a multi-layer CLIP feature extraction method with explicit pose-aware modulation. At CLIP layers l L 2, 4, 8, 12, features are extracted and concatenated with OpenPose keypoints of the reference human:

$$F_{sk} = \Sigma_l\ \alpha_l \cdot \mathrm{Conv}(\mathrm{PoseMap}(h)\ \odot\ F_l)$$

Learnable attention weights α z adaptively trade off coarse and fine-grained features with respect to the pose complexity. It is a design that can multitask in multi-person interactions, occlusions, and clothing detail - constraints that the base knowledge-enhanced GAN fails to do.

E. Preference-Optimized Objective

The generator loss combines four terms: adversarial loss Ladv of two discriminators (D1 and D2), CLIP alignment loss L_CLIP, human similarity loss L human-sim and the Reward Preference Optimization term L RPO:

$$L_G = L_{adv} + \lambda_1 L_{CLIP} + \lambda_2 L_{human\text{-}sim} + \lambda_3 L_{RPO}(\hat{r})$$

Here r is the estimated predicted rich feedback reward of a reward model that has been trained on RichHF-18K-style human preference annotations. The λ-Harmonic weighting schedule balances these terms dynamically during training, and speeds up convergence and enhances fine-grained semantic harmony in generated outputs.

IV. EXPERIMENTS

A. Datasets and Metrics

The contribution of each component is proved by systematic ablation. Eliminating dynamic multi-knowledge retrieval (replacing with top-1 based on training subset) reduces prompt consistency by 15%. Eliminating the RP loss makes identity preservation decrease by 9%. The elimination of the pose-aware aspect of AP-MDE leads to visible identity blending in multifaceted multi-person prompts. CMK-FT conflict mask in full form, does better by 18 points on FID than a naive concatenation baseline.

B. Quantitative Results — Single-Human Setting

TABLE I

Single-Human Results at 1024×1024 Resolution

Method	Images	Identity ↑	Prompt Cons. ↑	FID ↓	DreamSim ↓	Time (s)	Memory (GB)
Stable Diffusion	0	4.2	27.1	412	0.47	2.1	6.5
DreamBooth	5	28.5	24.3	385	0.52	950	38
FastComposer	1	52.3	25.1	355	0.24	2.0	6.2
IP-Adapter	1	62.1	27.4	298	0.21	3.8	6.0
Xu et al. (2025)	1	58.5	30.9	330	0.26	0.015	0.32
MK-POGAN (Ours)	**1**	**71.8**	**38.7**	**265**	**0.19**	**0.018**	**0.45**

Single-human results are shown in Table I. MK-POGAN has an identity preservation of 71.8, a 13.3 point higher than the best previous system IP-Adapter (62.1) and 22.7 points higher than Xu et al. (2025) (58.5) and has a prompt consistency score of 38.7, the highest of all baselines. FID improves to 265, compared to 298 for IP-Adapter and 330 for Xu et al. (2025). Most

importantly, the inference time is reduced to only 0.018s with 0.45 GB of memory, which is still approximately 300 times faster than diffusion-based models.

C. Quantitative Results — Multi-Human Setting

TABLE II

Multi-Human Results (2–4 Reference Humans per Prompt)

Method	Images	Identity ↑	Prompt Cons. ↑	FID ↓	DreamSim ↓	Time (s)	Memory (GB)
Stable Diffusion	0	2.1	28.9	481.4	0.51	2.3	7.1
Custom Diffusion	5	6.8	26.1	372.9	0.29	812	31
FastComposer	1	44.7	25.3	392.6	0.25	2.1	6.5
IP-Adapter	1	46.9	28.2	312.5	0.23	4.2	6.8
Xu et al. (2025)	1	45.2	30.1	329.8	0.26	0.015	0.32
MK-POGAN (Ours)	**1**	**52.8**	**36.4**	**278.3**	**0.18**	**0.021**	**0.52**

Multi-human generation results are reported in table II. MK-POGAN attains 52.8% identity preservation compared to the previous best (Xu et al., 2025), and has better prompt consistency (36.4 vs. 30.1) and lowest FID (278.3). The approach has similar inference speed (0.021s) even though it generates multiple subjects, which validates the scalability of the CMK-FT design.

D. Ablation Studies

Systematic ablation confirms the contribution of each component. Removing dynamic multi-knowledge retrieval (replacing with static top-1 from training subset) drops prompt consistency by 15%. Removing the RPO loss causes identity preservation to fall by 9%. Removing the pose-aware component of AP-MDE results in visible identity blending in complex multi-person prompts. The full CMK-FT conflict mask improves FID by 18 points over a naive concatenation baseline.

V. DISCUSSION

A. Knowledge Scalability

The contribution of each component is confirmed by systematic ablation. By eliminating dynamic multi-knowledge retrieval (instead of using static top-1 of training subset) prompt consistency decreases by 15%. Identity preservation decreases by 9% when the RP loss is removed. The elimination of the pose-aware aspect of AP-MDE leads to visible identity blending of complex multi-person prompts. The entire CMK-FT conflict mask raises FID by 18 points compared to a naive concatenation baseline.

B. Preference Alignment

The RPO aspect fills the gap in semantic harmony of previous GAN-based models. MK-POGAN is trained to predict human preferences by training a reward predictor on human preference annotations and using the predictor signal in the generator objective, thus learning to generate outputs that are socially similar to fine-grained human aesthetic preferences - especially in situations where there are multiple plausible generators. This is evidenced by the 28 percent consistent improvement in prompt and the high scores on DreamSim in all benchmarks.

C. Efficiency Analysis

The MK-POGAN is trained with under 6 percent of the Stable Diffusion parameters and, when run inference, is approximately 300 times faster, which is suitable for real-time usage. The small memory overhead over Xu et al. (2025) - 0.45 GB versus 0.32 GB - is due to the multi-knowledge fusion transformer, a small price to pay in the context of such high quality objectives. This makes MK-POGAN a viable choice to be used in production in personalized advertising, virtual try-on, and creative tools.

VI. FUTURE WORK

There are a number of directions to be explored. Video and 3D extension will demand temporal consistency measures in human-controlled video synthesis, in which frame-to-frame identity will be a non-trivial problem. Zero-shot multi-modal control may

combine inputs in the form of audio and pose to allow more controllability than by text prompts. Ethical protections such as bias reduction and deepfake detection modules are crucial to responsible deployment. Hybrid diffusion-GAN architectures can have the quality ceiling of diffusion and GAN efficiency. Lastly, privacy-preserving embeddings would enhance real-time web retrieval that would expand the knowledge base past pre-indexed databases.

VII. CONCLUSION

MK-POGAN, a Multi-Knowledge Augmented Preference-Optimized GAN, which was introduced in this paper, fills major gaps in human-controlled text-to-image synthesis: open-domain knowledge scalability, complex pose and detail fidelity, native high-resolution generation, and human preference alignment. MK-POGAN achieves state-of-the-art identity preservation (↑1218%), prompt consistency (↑28%), and FID scores on long FFHQ, COCO-Human, and Open-Human datasets, with 300 times faster inference than diffusion models using less than 6 percent of its parameters. These findings make MK-POGAN a valid, effective and quality solution to pragmatic open-domain creative generation. The release of code and models will be made on acceptance.

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